Q9. Multi-Container Nginx with Shared Volume — Solution

1) Create project folder

```
mkdir q9-nginx-shared
cd q9-nginx-shared
```

2) Create HTML file

Create a file named `index.html`

```
<!DOCTYPE html>
<html>
<head>
    <title>Nginx Shared Volume</title>
</head>
<body>
    <h1>Hello from Custom Nginx Image</h1>
    <p>This content is shared using Docker volume.</p>
</body>
</html>
```

3) Create Dockerfile

Create a file named `Dockerfile`

```
FROM nginx:latest
COPY index.html /usr/share/nginx/html/index.html
```

4) Build custom image

Replace `<username>` with your Docker Hub username.

```
docker build -t <username>/nginx-shared:v1 .
```

5) Create volume

```
docker volume create nginx-data-volume
```

6) Create network

```
docker network create nginx-network
```

7) Copy HTML into the shared volume using a temporary container

```
docker run --rm -v nginx-data-volume:/data -v $(pwd):/src u
buntu bash -c "cp /src/index.html /data/index.html"
```

8) Run first container

```
docker run -d \
--name nginx-app-1 \
--network nginx-network \
-v nginx-data-volume:/usr/share/nginx/html \
-p 8081:80 \
<username>/nginx-shared:v1
```

9) Run second container

```
docker run -d \
--name nginx-app-2 \
--network nginx-network \
-v nginx-data-volume:/usr/share/nginx/html \
-p 8082:80 \
<username>/nginx-shared:v1
```

10) Verify output

Open in browser:

```
http://localhost:8081
http://localhost:8082
```

Both should show the same page.

11) Verify shared volume change

Modify content in the volume:

```
docker run --rm -v nginx-data-volume:/data ubuntu bash -c
"echo '<h1>Updated Shared Content</h1>' > /data/index.html"
```

Refresh both pages:

```
http://localhost:8081
http://localhost:8082
```

Both containers should now show updated content.